Generative Modelling with Stochastic Differential Equations

Exercises, week 2

Langevin dynamics

Cristopher Salvi
Imperial College London

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3 Convergence of the Langevin diffusion and of ULA

Setting and notation. The following is used in all exercises of this chapter. $U \in C^2(\mathbb{R}^n)$ with $Z = \int e^{-U}\mathrm{d}x < \infty$, and $\mu(\mathrm{d}x) = Z^{-1}e^{-U(x)}\mathrm{d}x$ (we write μ also for its density). The *Langevin diffusion* is the solution of

$$\mathrm{d}X_t = -\nabla U(X_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t, \tag{LD}$$

with B an n-dimensional Brownian motion independent of X_0 (assumed to have a unique strong solution). Its semigroup is $P_t\varphi(x) = \mathbb{E}[\varphi(X_t)\,|\,X_0 = x]$ and its generator is $\mathcal{L}\varphi = \Delta\varphi - \langle\nabla U, \nabla\varphi\rangle$; we write $\mu_t = \mathrm{Law}(X_t)$, with density ρ_t. A probability measure ν is *invariant* if $\int P_t\varphi\,\mathrm{d}\nu = \int \varphi\,\mathrm{d}\nu$ for all bounded measurable φ and $t \geq 0$. With step size $h > 0$, the *unadjusted Langevin algorithm* (ULA) is

$$X_{k+1} = X_k - h\nabla U(X_k) + \sqrt{2h}\,\xi_k, \qquad \xi_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \mathbb{I}_n) \text{ independent of } X_0, \tag{ULA}$$

and $\mu^k := \mathrm{Law}(X_k)$. We use the assumptions

(A1) (strong convexity) $\nabla^2 U(x) \succeq \alpha\mathbb{I}_n$ for all x, for some $\alpha > 0$;

(A2) (smoothness) $-\beta\mathbb{I}_n \preceq \nabla^2 U(x) \preceq \beta\mathbb{I}_n$ for all x, i.e. ∇U is β-Lipschitz;

and under both we set $\kappa = \beta/\alpha \geq 1$ and let x^* be the minimiser of U. For $\nu \ll \mu$ with $f = \mathrm{d}\nu/\mathrm{d}\mu$ we use the divergences $\chi^2(\nu\|\mu) = \int(f-1)^2\mathrm{d}\mu$, $\mathrm{KL}(\nu\|\mu) = \int f\log f\,\mathrm{d}\mu$, $\mathrm{TV}(\nu,\mu) = \sup_A|\nu(A) - \mu(A)|$, and $\mathcal{W}_2(\nu,\mu)^2 = \inf_\pi \int|x-y|^2\mathrm{d}\pi(x,y)$ over couplings π of ν and μ. μ satisfies a *Poincaré inequality* with constant C if $\mathrm{Var}_\mu(\varphi) \leq C\int|\nabla\varphi|^2\mathrm{d}\mu$ for all smooth φ; the smallest such C is the Poincaré constant $C_{\mathrm{P}}(\mu)$. The log-Sobolev constant $C_{\mathrm{LSI}}(\mu)$ is the smallest C with $\mathrm{KL}(\nu\|\mu) \leq \frac{C}{2}\int|\nabla\log\frac{\mathrm{d}\nu}{\mathrm{d}\mu}|^2\mathrm{d}\nu$ for all smooth ν; one always has $C_{\mathrm{P}} \leq C_{\mathrm{LSI}}$. Finally, the following bound for ULA is proved in Exercise 3.7 (even with $\sqrt{3}$ replaced by 1):

(F) under (A1)–(A2) and $0 < h \leq 1/\beta$, for every μ^0 with finite second moment and all $k \geq 0$, $\mathcal{W}_2(\mu^k,\mu) \leq (1-\alpha h)^k\,\mathcal{W}_2(\mu^0,\mu) + \sqrt{3}\,\kappa\sqrt{nh}$.

Exercise 3.1 (Non-reversible Langevin dynamics)**.** Let J be a constant antisymmetric $n \times n$ matrix and consider $\mathrm{d}X_t = -(\mathbb{I}_n + J)\nabla U(X_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t$, whose generator is $\mathcal{L}_J\varphi = \mathcal{L}\varphi - \langle J\nabla U, \nabla\varphi\rangle$.

(a) Show that $\int \mathcal{L}_J\varphi\,\mathrm{d}\mu = 0$ for $\varphi \in C_c^\infty(\mathbb{R}^n)$, so that (under the same regularity as for the Langevin diffusion) μ is still invariant.

(b) More generally, show that $\int\langle b, \nabla\varphi\rangle\,\mathrm{d}\mu = 0$ for all $\varphi \in C_c^\infty(\mathbb{R}^n)$ whenever b is a smooth vector field with $\nabla\cdot(b\mu) = 0$, so that every drift $-\nabla U + b$ of this kind leaves μ invariant.

Exercise 3.2 (Strong convexity, monotonicity and uniqueness)**.** Let $U \in C^2(\mathbb{R}^n)$ satisfy (A1).

(a) Show that $U(y) \geq U(x) + \langle\nabla U(x), y-x\rangle + \frac{\alpha}{2}|y-x|^2$ for all x, y. Deduce that U has a unique minimiser x^* and that $\int e^{-U} < \infty$.

(b) Assume moreover (A2). Show that for all $h > 0$,

$$|x - h\nabla U(x) - (y - h\nabla U(y))| \le \max\{|1 - h\alpha|, |1 - h\beta|\}\,|x - y|,$$

and that the factor equals $1 - \alpha h$ when $0 < h \le 2/(\alpha + \beta)$, in particular when $h \le 1/\beta$.

(c) Show that μ is the unique invariant probability measure of the Langevin diffusion (no moment assumption on the invariant measure).

Hint for (c): run the synchronous coupling of Theorem 3.4 from independent $X_0 \sim \nu$ and $Y_0 \sim \mu$, where ν is invariant, and compare $\int \varphi \,\mathrm{d}\nu$ with $\int \varphi \,\mathrm{d}\mu$ for bounded Lipschitz φ.

Exercise 3.3 (Reversibility of the Langevin diffusion)**.** Let $\mathcal{A}$ be a class of bounded smooth functions with bounded derivatives, stable under P_t, such that $\partial_s P_s\varphi = \mathcal{L}P_s\varphi$ for $\varphi \in \mathcal{A}$ (with differentiation under the integral sign justified) and such that the symmetry identity

$$\int \varphi\, \mathcal{L}\psi \,\mathrm{d}\mu = -\int \langle \nabla\varphi, \nabla\psi\rangle \,\mathrm{d}\mu$$

holds for all $\varphi, \psi \in \mathcal{A}$.

(a) Show that P_t is symmetric in $L^2(\mu)$: $\int \varphi\, P_t\psi \,\mathrm{d}\mu = \int \psi\, P_t\varphi \,\mathrm{d}\mu$ for $\varphi, \psi \in \mathcal{A}$.

(b) Let Y be the Langevin diffusion with $Y_0 \sim \mu$. Deduce that (Y_0, Y_t) and (Y_t, Y_0) have the same law (reversibility), assuming $\mathcal{A}$ is rich enough that products $\varphi(x)\psi(y)$, $\varphi, \psi \in \mathcal{A}$, determine laws on $\mathbb{R}^n \times \mathbb{R}^n$.

Exercise 3.4 (Sharpness of the χ^2 bound on total variation)**.** Recall that $\mathrm{TV}(\nu, \mu) \le \frac{1}{2}\sqrt{\chi^2(\nu\|\mu)}$ for all probability measures ν, μ.

(a) Show that equality holds for $\nu = \mathrm{Bernoulli}(\frac{1}{2} + \varepsilon)$ and $\mu = \mathrm{Bernoulli}(\frac{1}{2})$.

(b) If μ satisfies a Poincaré inequality with constant C_{P}, the Langevin diffusion satisfies $\chi^2(\mu_t\|\mu) \le e^{-2t/C_{\mathrm{P}}}\chi^2(\mu_0\|\mu)$. Deduce a bound on $\mathrm{TV}(\mu_t, \mu)$, and evaluate it for $\mu = \gamma_n$ and $\mu_0 = \mathcal{N}(m, \mathbb{I}_n)$. How long must one wait for the bound to be small when $|m| \approx \sqrt{n}$?

Exercise 3.5 (Bottlenecks force a large Poincaré constant)**.** Let μ be a probability measure on $\mathbb{R}^n$ and e a unit vector. Only the coordinate $\langle e, x\rangle$, in the direction separating the modes, will matter. Write $A = \{\langle e, x\rangle \le -1\}$, $B = \{|\langle e, x\rangle| < 1\}$, $C = \{\langle e, x\rangle \ge 1\}$, and let $\varphi(x) = 1$ on A, $\varphi(x) = -\langle e, x\rangle$ on B, $\varphi(x) = -1$ on C. You may use the Poincaré inequality for Lipschitz functions, with $\nabla\varphi$ the a.e. gradient.

(a) Show that $|\mathrm{Var}_\mu(\varphi) - 4\mu(A)\mu(C)| \le 5\mu(B)$, and deduce $C_{\mathrm{P}}(\mu) \ge (4\mu(A)\mu(C) - 5\mu(B))/\mu(B)$.

(b) Let $\mu_m = p\,\mathcal{N}(-me, \mathbb{I}_n) + (1-p)\,\mathcal{N}(me, \mathbb{I}_n)$ with $p \in (0, 1)$ and $m \ge 1$. Show that

$$C_{\mathrm{P}}(\mu_m) \ge p(1-p)\sqrt{\pi/2}\; e^{(m-1)^2/2} - 5.$$

(c) Interpret (b), recalling that under a Poincaré inequality the Langevin diffusion satisfies $\chi^2(\mu_t\|\mu) \le e^{-2t/C_{\mathrm{P}}}\chi^2(\mu_0\|\mu)$, and that each mixture component $\mathcal{N}(\pm me, \mathbb{I}_n)$ has Poincaré constant 1.

Exercise 3.6 (Moment bounds for μ)**.** Assume (A1) and (A2), and let x^* be the minimiser of U. Lemma 3.12 proves $\mathbb{E}_\mu|\nabla U|^2 = \mathbb{E}_\mu \Delta U \le n\beta$ and $\mathbb{E}_\mu|X - x^*|^2 \le n/\alpha$ from the identity $\mathbb{E}_\mu \nabla\cdot F = \mathbb{E}_\mu\langle F, \nabla U\rangle$. Show that $\mathbb{E}_\mu|\nabla U|^2 \ge n\alpha$ and $\mathbb{E}_\mu|X - x^*|^2 \ge n/\beta$, and that all four bounds are attained by Gaussian measures.

Exercise 3.7 (Short-time displacement and ULA in $\mathcal{W}_2$)**.** Assume (A1) and (A2), and let Y solve (LD) with $Y_0 \sim \mu$. You may use that μ is invariant, so that $Y_t \sim \mu$ for all t, and that $\mathbb{E}_\mu|\nabla U|^2 \le n\beta$ (Lemma 3.12).

(a) Show that $\mathbb{E}|Y_{t+\tau} - Y_t|^2 \le 2n\beta\tau^2 + 4n\tau$ for all $t, \tau \ge 0$.

(b) Using Itô's formula for $|Y_t - Y_0|^2$ and reversibility (Exercise 3.3(b)), show the sharper bound $\mathbb{E}|Y_{t+\tau} - Y_t|^2 \le 2n\tau$. (Hint: Itô's formula leaves the term $-2\int_0^\tau \mathbb{E}\langle Y_t - Y_0, \nabla U(Y_t)\rangle \,\mathrm{d}t$. Use reversibility to rewrite $\mathbb{E}\langle Y_t - Y_0, \nabla U(Y_t)\rangle$ with Y_0 and Y_t exchanged, average the two expressions, and use the monotonicity of ∇U.)

(c) Let $0 < h \le 1/\beta$. Let (X_0, Y_0) be a coupling of μ^0 and μ, independent of B, and run ULA (ULA) with $\xi_k := h^{-1/2}(B_{(k+1)h} - B_{kh})$. Show that $w_k := (\mathbb{E}|X_k - Y_{kh}|^2)^{1/2}$ satisfies $w_{k+1} \le (1-\alpha h)w_k + \beta\sqrt{n}\, h^{3/2}$. (Hint: split $X_{k+1} - Y_{(k+1)h}$ into a gradient-descent term, contracted by Exercise 3.2(b), and a discretisation term $\int_{kh}^{(k+1)h}(\nabla U(Y_s) - \nabla U(Y_{kh}))\mathrm{d}s$.)

(d) Deduce that $\mathcal{W}_2(\mu^k, \mu) \le (1-\alpha h)^k \mathcal{W}_2(\mu^0, \mu) + \kappa\sqrt{nh}$, and that using (a) instead of (b) gives (F).

(e) For the OU process ($U = |x|^2/2$), compute $\mathbb{E}|Y_\tau - Y_0|^2$ exactly and compare with (b).

Exercise 3.8 (Uniqueness of the invariant measure of ULA)**.** Assume (A1) and (A2) and let $0 < h \le 1/\beta$. Proposition 3.11 shows that ULA has a unique invariant measure μ_h with finite second moment.

(a) Show that μ_h is the unique invariant probability measure of ULA, even among measures without a second moment.

(b) Deduce $\mathcal{W}_2(\mu_h, \mu) \le \sqrt{3}\,\kappa\sqrt{nh}$ from (F).

Hint for (a): argue as in Exercise 3.2(c), with two ULA chains driven by the same noise.

Exercise 3.9 (Iteration complexity of ULA)**.** Assume (A1) and (A2), run ULA from $X_0 = x^*$ (the minimiser of U) and let $\varepsilon \in (0, 1]$.

(a) Show that with $h = \frac{\varepsilon^2}{12n\beta\kappa}$ and $k \ge \frac{12\kappa^2 n}{\varepsilon^2}\log\frac{2\sqrt{n}}{\varepsilon}$, one has $\mathcal{W}_2(\mu^k, \mu) \le \varepsilon/\sqrt{\alpha}$. (Use (F) and Lemma 3.12.)

(b) Why is $\varepsilon/\sqrt{\alpha}$ (rather than ε) the natural accuracy scale?

(c) What changes if one starts from a point x_0 with $|x_0 - x^*| \le R$?

Exercise 3.10 (ULA with inexact gradients)**.** Assume (A1) and (A2), and let $G : \mathbb{R}^n \to \mathbb{R}^n$ be measurable with $|G(x) - \nabla U(x)| \le \delta$ for all x, and consider $X_{k+1} = X_k - hG(X_k) + \sqrt{2h}\,\xi_k$ with $0 < h \le 1/\beta$.

(a) Show that $\mathcal{W}_2(\mathrm{Law}(X_k), \mu) \le (1-\alpha h)^k \mathcal{W}_2(\mathrm{Law}(X_0), \mu) + \sqrt{3}\,\kappa\sqrt{nh} + \delta/\alpha$. (Hint: adapt the coupling of Exercise 3.7(c).)

(b) Explain why this bias does not vanish as $h \to 0$, and why a uniform bound on the gradient error is an unrealistic assumption for a learned score.